



The 6th International Conference on Power Electronics and their Applications

Design and Evaluation of a Double-Loop ADRC Scheme

FOR PUSH–PULL CONVERTERS

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Presentation Outline

- ▶ System Description and Modeling
- ▶ Fundamentals of First-Order ADRC
- ▶ Double-Loop ADRC Architecture
- ▶ Simulation Results and Performance Evaluation



Presentation Outline

1 System Description and Modeling

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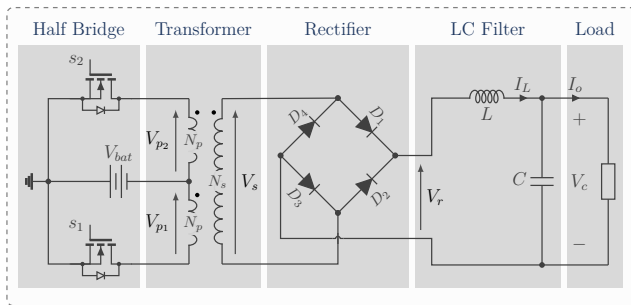


Description

1 System Description and Modeling

The push-pull converter is an isolated DC-DC converter made of four stages:

- a half-bridge,
- a transformer,
- a rectifier,
- and an LC filter.



■ This structure provides **galvanic isolation**, **efficient power transfer**, and **flexible voltage scaling**.



Description

1 System Description and Modeling

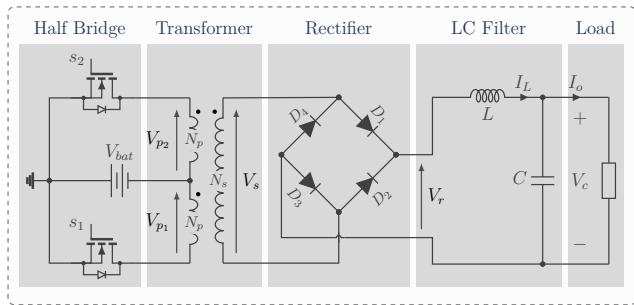


Figure: Circuit Diagram of a Push-Pull Converter.

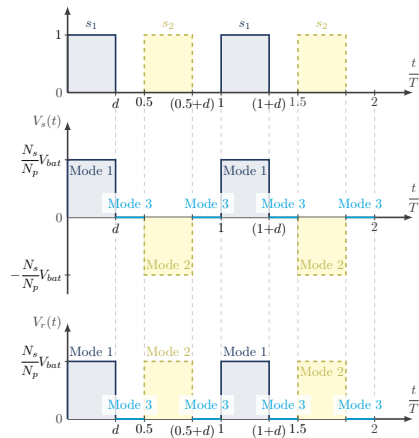


Figure: Switching stage waveforms.



Description

1 System Description and Modeling

- ❶ Similar structure to a buck converter.
- ❷ Effective switching frequency is doubled ($2f_{sw}$).
- ❸ Better ripple attenuation, simplifying filter design.
- ❹ Transformer provides voltage scaling and galvanic isolation.

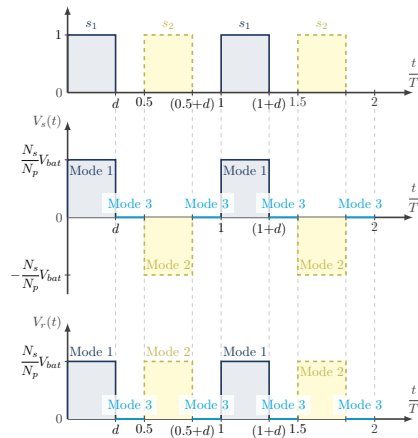


Figure: Switching stage waveforms.



Mathematical Model

1 System Description and Modeling

The average model of the push-pull converter is given by:

$$\frac{dI_L(t)}{dt} = \frac{2}{L} \frac{N_s}{N_p} V_{bat} U(t) - \frac{1}{L} V_c(t) \quad (1)$$

$$\frac{dV_c(t)}{dt} = \frac{1}{C} I_L(t) - \frac{1}{C} I_o(t) \quad (2)$$

- $I_L(t)$ and $V_c(t)$ represent the average **inductor current** and **capacitor voltage**, respectively, over a PWM switching period.
 - $U(t)$ is the **control** variable corresponding to the duty cycle D , and $I_o(t)$ denotes the **unknown load current**.
-



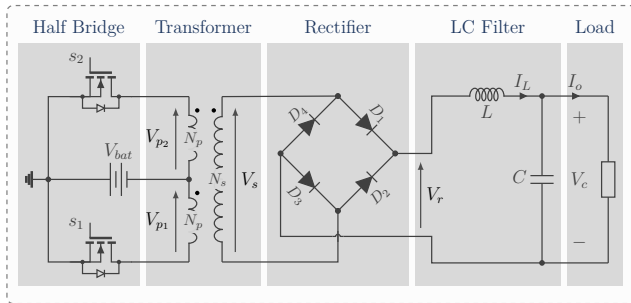
Control Objectives

1 System Description and Modeling

■ **Regulate** the output voltage V_c around the reference V_{ref}

■ Ensure **robustness** against:

- **Load variations** $I_o(t)$,
- **Input-voltage fluctuations** V_{bat} ,
- **Parameter uncertainties.**





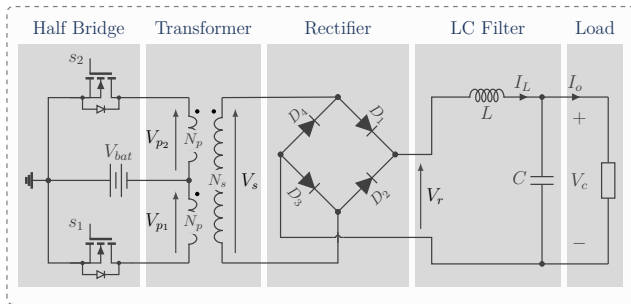
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Active Disturbance Rejection Control

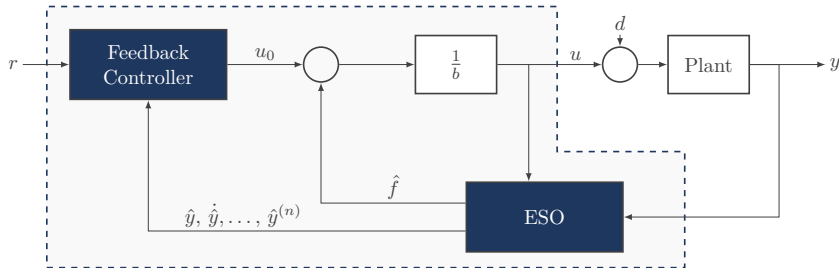


Controller Choice and Justification

1 System Description and Modeling

ADRC** simplifies the system control by combining all internal uncertainties and external disturbances into one **generalized disturbance**.

An Extended State Observer (ESO) estimates this disturbance in real time, which allows the controller to actively cancel its effect.



**Jingqing Han, "From PID to Active Disturbance Rejection Control," *IEEE Transactions on Industrial Electronics*, vol. 56, no. 3, pp. 900–906, 2009.



Presentation Outline

2 Fundamentals of First-Order ADRC

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First-Order ADRC Formulation

2 Fundamentals of First-Order ADRC

The first-order ADRC treats first-order plants as a disturbed integrator:

$$\dot{y}(t) = f(t, y, u) + b_0 u(t) \quad (3)$$

where:

- $y(t)$ is the system output,
- $u(t)$ is the control input,
- b_0 is the nominal control gain,
- $f(t, y, u)$ represents the total disturbance, assumed **differentiable and bounded**.



Extended State-Space Model

2 Fundamentals of First-Order ADRC

Consider the states $x_1 = y$ and $x_2 = f$, the system can be re-expressed as:

$$\begin{cases} \dot{x}_1(t) = x_2(t) + b_0 u(t) \\ \dot{x}_2(t) = \dot{f}(t) \end{cases} \quad (4)$$

and in a compact state-space form as:

$$\dot{x} = Ax(t) + Bu(t) + Eh(t) \quad (5)$$

with matrices:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} b_0 \\ 0 \end{bmatrix}, \quad E = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (6)$$

where $h = \dot{f}$ denotes the disturbance dynamics.



ESO-Based Disturbance Estimation

2 Fundamentals of First-Order ADRC

To estimate the output x_1 and disturbance x_2 , we use a full-order Luenberger observer:

$$\dot{\hat{x}} = A\hat{x} + Bu + L_o(x_1 - \hat{x}_1)$$

with the observer gain

$$L_o = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}$$

When the observer is stable, $\hat{x}_1$ tracks the output and $\hat{x}_2$ estimates the total disturbance.



ADRC-Based Control Design

2 Fundamentals of First-Order ADRC

With the following control expression:

$$u = \frac{1}{b_0} (-\hat{x}_2 + u_o) \quad (7)$$

The first-order disturbed system becomes:

$$\dot{y}(t) = f(t, y, u) + b_0 \frac{1}{b_0} (-\hat{x}_2 + u_o)$$



ADRC-Based Control Design

2 Fundamentals of First-Order ADRC

With the following control expression:

$$u = \frac{1}{b_0} (-\hat{x}_2 + u_o) \quad (7)$$

The first-order disturbed system becomes:

$$\dot{y}(t) = f(t, y, u) + b_0 \frac{1}{b_0} (-\hat{x}_2 + u_o) \implies \dot{y}(t) = f - \hat{x}_2 + u_o \quad (8)$$

If the observer is properly tuned such that $\hat{x}_2 \approx f$, then the system reduces to:

$$\dot{y} \approx u_o \quad (9)$$

The reduced system can be controlled by a proportional state-feedback law:

$$u_o = k_p (r - x_1) \quad (10)$$

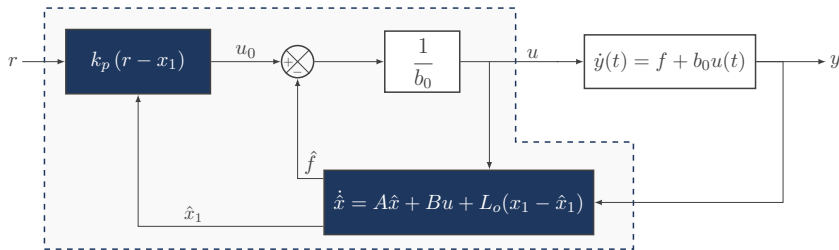


ADRC: Key Steps

2 Fundamentals of First-Order ADRC

The ADRC design can be summarized in four main steps:

1. Extended State-Space Model
2. Extended State Observer (ESO)
3. Disturbance Cancellation
4. Proportional Gain Design





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Application to the Push-Pull Converter Model

3 Double-Loop ADRC Architecture

The average model of the push-pull converter is given by:

$$\frac{dI_L(t)}{dt} = \frac{2}{L} \frac{N_s}{N_p} V_{bat} U(t) - \frac{1}{L} V_c(t), \quad \frac{dV_c(t)}{dt} = \frac{1}{C} I_L(t) - \frac{1}{C} I_o(t)$$

The inductor current can be expressed as a disturbed integrator system:

$$\dot{y}_I(t) = b_I U(t) + f_I(t) \quad (11)$$

where $y_I(t) = I_L(t)$ is the system output, $b_I = \frac{2}{L} \frac{N_s}{N_p} V_{bat}^0$ is the nominal control gain, and the total disturbance:

$$f_I(t) = \frac{2}{L} \frac{N_s}{N_p} \delta V_{bat} U(t) - \frac{1}{L} V_c(t) \quad (12)$$

The capacitor voltage can be reformulated as a disturbed integrator system:

$$\dot{y}_V(t) = b_V u(t) + f_V(t) \quad (13)$$

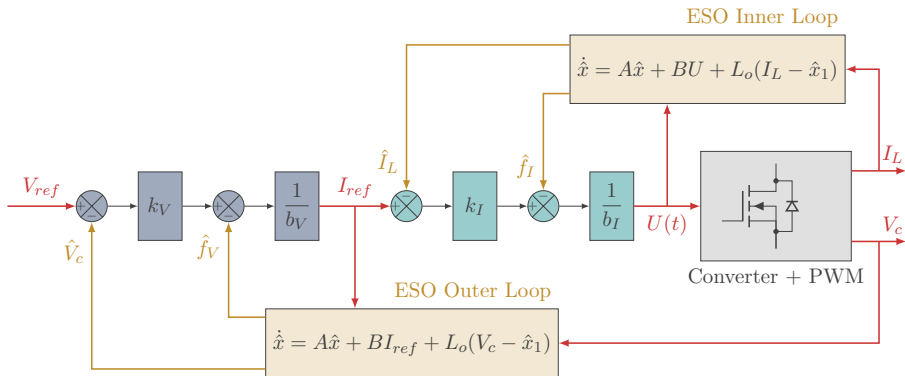
where $y_V(t) = V_c(t)$ is the system output, $u(t) = I_L(t)$ which corresponds to the reference current, $b_V = \frac{1}{C}$ is the control gain, and the total disturbance:

$$f_V(t) = -\frac{1}{C} I_o(t) \quad (14)$$



Application to the Push-Pull Converter Model

3 Double-Loop ADRC Architecture





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System and Controller Parameters

4 Simulation Results and Performance Evaluation

The double-loop ADRC was tested on a typical push-pull converter with an input voltage of $V_{bat}^0 = 24V$, a filter inductance of $L = 1.57mH$, and a filter capacitance of $C = 220\mu F$.

The high frequency transformer has a turns ratio of $N_s/N_p = 25$ and operated at a PWM switching frequency of 30kHz.

The double-loop ADRC parameters are chosen as follows:

- For the inner current loop, the controller bandwidth was set to $\omega_{c1} = 2\pi \times 1000$ rad/s, and the observer bandwidth to $\omega_{o1} = 3\omega_{c1}$.
- For the outer voltage loop, the controller bandwidth was $\omega_{c2} = 2\pi \times 100$ rad/s, and the observer bandwidth $\omega_{o2} = 3\omega_{c2}$.



Disturbance profiles

4 Simulation Results and Performance Evaluation

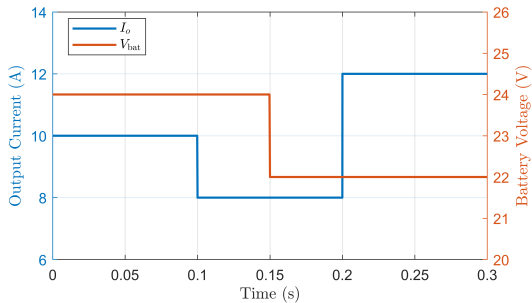


Figure: Applied load current and input voltage disturbance profiles.



Tracking performance

4 Simulation Results and Performance Evaluation

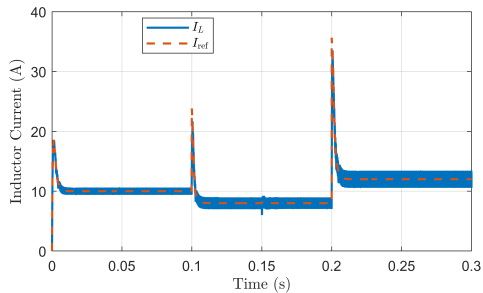
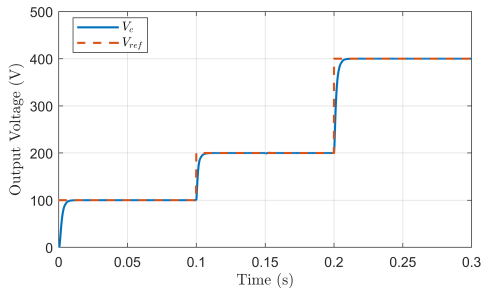


Figure: Output voltage and inductor current trackings performance under load and input variations.



Disturbance Estimation

4 Simulation Results and Performance Evaluation

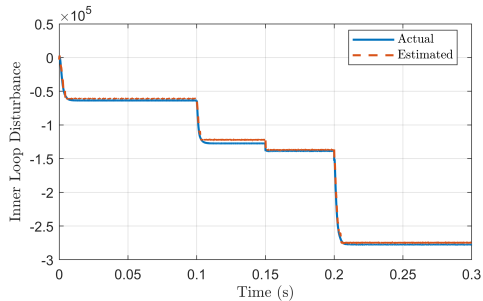
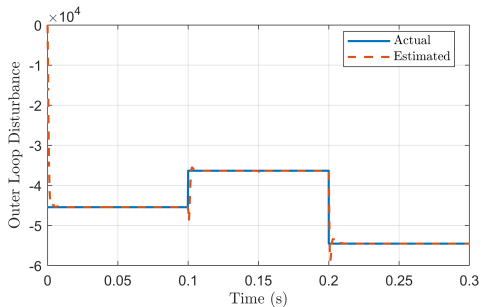


Figure: Inner loop and Outer loop disturbances estimation.



Conclusion & Future Works

- The double-loop ADRC achieved robust voltage regulation under load and input disturbances.
- The controller provided fast, accurate tracking with zero steady-state error, even during abrupt changes.
- Disturbance estimation in both loops closely matched the applied disturbances, enabling precise compensation.



Q&A

Thank you for listening!